



Know your appetite: Phosphate-sensing proteins regulate AM symbiosis

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We often face the challenge of deciding when and what to eat, and so do plants. The decisions plants make regarding the timing, amount, and type of nutrient to consume are entangled with their nutrient status, the amount of freely available nutrients in the soil, and their interactions with nutrient-providing microbes, such as arbuscular mycorrhizal (AM) fungi. AM fungi, which have partnered with plants since they first came on land over 400 million years ago, provide inorganic phosphate (Pi) and other mineral nutrients to plants in exchange for photosynthetically fixed carbon in the form of sugars and lipids (Chiu and Paszkowski, 2019). Nutrient exchange occurs at an ephemeral, tree-shaped symbiotic interface in root cortical cells called the arbuscule (see Figure), whose formation is strictly controlled by the demand of Pi by the plant. Plants can also monitor Pi status after the initiation of AM symbiosis and supply more carbon to AM fungi that deliver more Pi, a phenomenon called reciprocal rewarding (Kiers et al., 2011). Therefore, the existence of a Pi-monitoring system that regulates AM symbiosis has been postulated but molecularly uncharacterized.

In eukaryotes, cellular Pi status is sensed by SPX domain-containing proteins (SPXs) through their interactions with inositol polyphosphates (PP-InsPs), whose abundance is altered by cellular Pi concentration (Jung et al., 2018). In *Arabidopsis* (*Arabidopsis thaliana*) and rice (*Oryza sativa*), SPXs bound by PP-InsPs inactivate Pi starvation response (PSR) transcription factors under Pi-sufficient conditions via direct interactions. Although the role of SPXs in Pi homeostasis under non-symbiotic conditions is well characterized, whether Pi-sensing proteins play roles during AM symbiosis has been unknown. In a new study, Peng Wang and colleagues (Wang et al., 2021) demonstrate that *Medicago*

truncatula SPXs regulate multiple steps in AM symbiosis—from birth to death of arbuscules—as well as Pi homeostasis under non-mycorrhizal conditions.

The authors first showed that, among the six SPX genes in *M. truncatula*, SPX1 and SPX3 were strongly induced in multiple root cell types under Pi starvation in the absence of AM fungi. Consistent with the role of SPX proteins in *Arabidopsis* and rice, SPX1 and SPX3 interact with the PSR transcription factor PHR2 in a Pi concentration-dependent manner and inhibit its activity to control PSRs and Pi homeostasis.

Interestingly, the expression of SPX1 and SPX3 was dramatically altered and restricted in arbuscule-containing cells during AM symbiosis (see Figure). Single and double mutants of SPX1 and SPX3 showed significantly reduced colonization of AM fungi, indicating that the SPXs positively regulate AM colonization. In addition, SPX1 and SPX3 positively and redundantly regulate arbuscule degradation, a less understood process in AM symbiosis. The *spx1 spx3* double mutant showed reduced expression of MYB1, a known regulator of arbuscule degradation; no interaction between the SPX proteins and MYB1 was observed. How SPX1 and SPX3 regulate arbuscule degradation requires further investigation.

The authors showed that, under low Pi conditions, SPX1 and SPX3 positively regulate the expression of D27, which controls the biosynthesis of strigolactones, a class of plant hormones that also regulate the growth and branching of AM fungi. Thus, SPX1 and SPX3 may control AM colonization by regulating the production of strigolactones. In support of this hypothesis, root exudates of *spx* mutants failed to induce branching in AM fungi. Further analysis of root exudates, including direct measurement of strigolactones,